

6-data_preprocessing_with_column_transformer

October 22, 2024

1 Data Preprocessing With Column Transformer

The ColumnTransformer in Python's sklearn library is a powerful tool that allows you to apply different preprocessing steps to different columns of your dataset. This is particularly useful when you have a mixture of numerical and categorical data and need to apply distinct transformations to each type of data. Using the ColumnTransformer, you can combine transformations like scaling numerical features, encoding categorical variables, and imputing missing values, all in a single unified process.

Why Use ColumnTransformer?

1. **Multiple Data Types:** It handles datasets with mixed data types (numerical, categorical, text, etc.).
2. **Efficient Pipeline:** You can combine different preprocessing steps into a pipeline, making the code cleaner and easier to maintain.
3. **Selective Preprocessing:** Apply different preprocessing steps to specific columns, instead of preprocessing the entire dataset uniformly.
4. **Integration with Pipelines:** It can be combined with machine learning models in a Pipeline, allowing for a streamlined workflow from preprocessing to model training and evaluation.

```
[2]: import pandas as pd

d = {'sales': [
    100000, 222000, 1000000, 522000, 111111, 222222, 1111111, 20000, 75000, 90000, 1000000, 10000],
     'city': [
    'Tampa', 'Tampa', 'Orlando', 'Jacksonville', 'Miami', 'Jacksonville', 'Miami', 'Miami', 'Orlando',
    'Medium', 'Large', 'Large', 'Small', 'Medium', 'Large', 'Small', 'Medium', 'Medium', 'Medium', 'Small']

df = pd.DataFrame(data=d)
df
```

```
[2]:      sales      city      size
0   100000     Tampa   Small
1   222000     Tampa  Medium
2  1000000   Orlando   Large
```

3	522000	Jacksonville	Large
4	111111	Miami	Small
5	222222	Jacksonville	Medium
6	111111	Miami	Large
7	20000	Miami	Small
8	75000	Orlando	Medium
9	90000	Orlando	Medium
10	1000000	Orlando	Medium
11	10000	Orlando	Small

```
[7]: from sklearn.preprocessing import OneHotEncoder, OrdinalEncoder
from sklearn.compose import make_column_transformer

ohe = OneHotEncoder(sparse_output=False)
ode = OrdinalEncoder()

ct = make_column_transformer(
    (ohe, ['city']),
    (ode, ['size']),
    remainder='passthrough'
)

ct.set_output(transform='pandas')
```

```
[7]: ColumnTransformer(remainder='passthrough',
                        transformers=[('onehotencoder',
                                      OneHotEncoder(sparse_output=False), ['city']),
                                      ('ordinalencoder', OrdinalEncoder(), ['size'])])
```

```
[8]: df_pandas = ct.fit_transform(df)
df_pandas
```

```
[8]:
```

	onehotencoder__city_Jacksonville	onehotencoder__city_Miami	\
0	0.0	0.0	
1	0.0	0.0	
2	0.0	0.0	
3	1.0	0.0	
4	0.0	1.0	
5	1.0	0.0	
6	0.0	1.0	
7	0.0	1.0	
8	0.0	0.0	
9	0.0	0.0	
10	0.0	0.0	
11	0.0	0.0	

	onehotencoder__city_Orlando	onehotencoder__city_Tampa	\
--	-----------------------------	---------------------------	---

0	0.0	1.0
1	0.0	1.0
2	1.0	0.0
3	0.0	0.0
4	0.0	0.0
5	0.0	0.0
6	0.0	0.0
7	0.0	0.0
8	1.0	0.0
9	1.0	0.0
10	1.0	0.0
11	1.0	0.0

	ordinalencoder__size	remainder__sales
0	2.0	100000
1	1.0	222000
2	0.0	1000000
3	0.0	522000
4	2.0	111111
5	1.0	222222
6	0.0	1111111
7	2.0	20000
8	1.0	75000
9	1.0	90000
10	1.0	1000000
11	2.0	10000

```
[9]: ct2 = make_column_transformer(
      (ohe, [1]),
      (ode, [2]),
      remainder='drop'
    )

    ct2.set_output(transform='pandas')
```

```
[9]: ColumnTransformer(transformers=[('onehotencoder',
                                      OneHotEncoder(sparse_output=False), [1]),
                                      ('ordinalencoder', OrdinalEncoder(), [2])])
```

```
[11]: df_pandas2 = ct2.fit_transform(df)
      df_pandas2
```

```
[11]: onehotencoder__city_Jacksonville  onehotencoder__city_Miami  \
0                                     0.0                      0.0
1                                     0.0                      0.0
2                                     0.0                      0.0
3                                     1.0                      0.0
```

4	0.0	1.0
5	1.0	0.0
6	0.0	1.0
7	0.0	1.0
8	0.0	0.0
9	0.0	0.0
10	0.0	0.0
11	0.0	0.0

	onehotencoder__city_Orlando	onehotencoder__city_Tampa \
0	0.0	1.0
1	0.0	1.0
2	1.0	0.0
3	0.0	0.0
4	0.0	0.0
5	0.0	0.0
6	0.0	0.0
7	0.0	0.0
8	1.0	0.0
9	1.0	0.0
10	1.0	0.0
11	1.0	0.0

	ordinalencoder__size
0	2.0
1	1.0
2	0.0
3	0.0
4	2.0
5	1.0
6	0.0
7	2.0
8	1.0
9	1.0
10	1.0
11	2.0

```
[12]: ct3 = make_column_transformer(
      (ohe, [1]),
      ('passthrough', ['size']),
      remainder='drop'
    )

    ct3.set_output(transform='pandas')
```

```
[12]: ColumnTransformer(transformers=[('onehotencoder',
                                       OneHotEncoder(sparse_output=False), [1]),
```

```
('passthrough', 'passthrough', ['size']))
```

```
[13]: ct3.fit_transform(df)
```

```
[13]:
```

	onehotencoder__city_Jacksonville	onehotencoder__city_Miami	\
0	0.0	0.0	
1	0.0	0.0	
2	0.0	0.0	
3	1.0	0.0	
4	0.0	1.0	
5	1.0	0.0	
6	0.0	1.0	
7	0.0	1.0	
8	0.0	0.0	
9	0.0	0.0	
10	0.0	0.0	
11	0.0	0.0	

	onehotencoder__city_Orlando	onehotencoder__city_Tampa	passthrough__size
0	0.0	1.0	Small
1	0.0	1.0	Medium
2	1.0	0.0	Large
3	0.0	0.0	Large
4	0.0	0.0	Small
5	0.0	0.0	Medium
6	0.0	0.0	Large
7	0.0	0.0	Small
8	1.0	0.0	Medium
9	1.0	0.0	Medium
10	1.0	0.0	Medium
11	1.0	0.0	Small